MH1101 Tutorial 4 (Week 5)

Solution

Reference: Sections 2.3, 3.1 (Lecture Notes)

1. Use the method of cylindrical shells to find the volume generated by rotating the region bounded by the given curves about the given axis.

(i) $y = x^2$, $y = 6x - 2x^2$; about $x = 0$.

(ii) $y = x^3$, $y = 8$, $x = 0$; about $x = 3$.

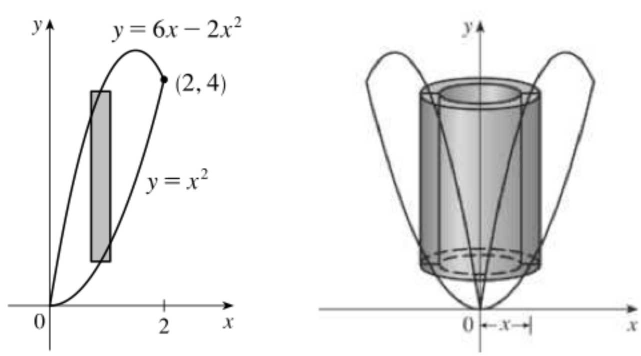
(iii) $x = 2y^2$, $x = y^2 + 1$; about $y = -2$.

Solution

(i) Intersection points: $x^2 = 6x - 2x^2 \iff 3x^2 - 6x = 0 \iff 3x(x-2) = 0 \iff x = 0,\ x = 2$. The intersection points are $(0,0)$, $(2,4)$.

A typical shell has

- radius x (rotation about y-axis), and so its circumference is $2\pi x$,
- height $(6x - 2x^2) - x^2$.

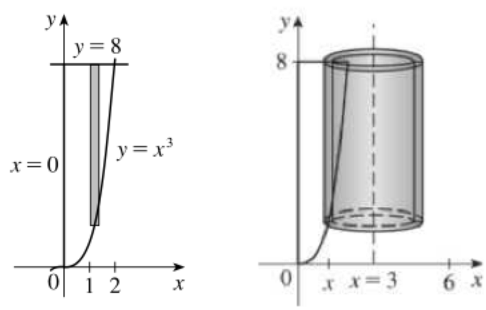


The volume is

$$\begin{aligned} V &= \int_0^2 2\pi x\left((6x-2x^2)-x^2\right)\,dx = 2\pi\int_0^2 (6x^2-3x^3)\,dx \\ &= 2\pi\left[2x^3-\frac{3x^4}{4}\right]_0^2 = 2\pi\,(16-12) = 8\pi. \end{aligned}$$

(ii) A typical shell has

- radius $3-x$ (rotation about $x=3$), so its circumference is $2\pi(3-x)$.
- height $8-x^3$.



The volume is

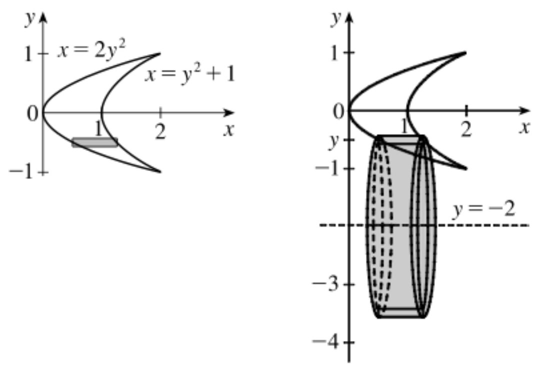
$$\begin{aligned} V &= \int_0^2 2\pi(3-x)(8-x^3)\,dx = 2\pi\int_0^2 (24-3x^3-8x+x^4)\,dx \\ &= 2\pi\left[24x-\frac{3x^4}{4}-4x^2+\frac{x^5}{5}\right]_0^2 = 2\pi\left(48-12-16+\frac{32}{5}\right) = \frac{264}{5}\pi. \end{aligned}$$

(iii) Intersection points: $2y^2 = y^2+1 \implies y^2 = 1 \implies y = \pm 1$. So the the intersection points of the two curves are $(2,1)$, $(2,-1)$.

A typical shell has

- radius $(y-(-2)) = y+2$ (rotation about $y=-2$), so its circumference is $2\pi(y+2)$,

– height $(y^2+1) - 2y^2 = 1 - y^2$.



The volume is

$$
\begin{aligned}
V &= \int_{-1}^{1} 2\pi(y+2)(1-y^2)\,dy = 2\pi \int_{-1}^{1} (y - y^3 + 2 - 2y^2)\,dy \\
&= 2\pi \left[\frac{y^2}{2} - \frac{y^4}{4} + 2y - \frac{2y^3}{3} \right]_{-1}^{1} = \frac{16}{3}\pi.
\end{aligned}
$$

□

2. Each integral represents the volume of a solid. Describe the solid.

(i) $\displaystyle\int_0^3 2\pi x^5\,dx.$

(ii) $\displaystyle\int_0^{\pi/2} 2\pi(x+1)(2x - \sin x)\,dx.$

Solution

For this type of question, answers may not be unique; it depends on how you identify the various curves bounding the region, and also whether you express the volume using disk-washer or cylindrical shells. For the solution below, we will use cylindrical shells.

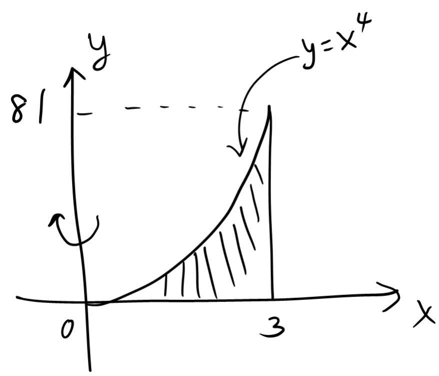
(i)

$$\int_0^3 2\pi x^5\,dx = \int_0^3 \underbrace{(2\pi x)}_{\text{circumference}}\ \underbrace{(x^4)}_{\text{height}}\,dx.$$

The radius of the shell is x, so the solid is rotated about the y-axis.

Its height is x^4, so the solid is obtained by rotating the region bounded between $y = x^4$, $y = 0$, $x = 3$.

From the limits of integration, we conclude that this is the solid obtained by rotating the region bounded by the curves $y = x^4$ and $y = 0$, $x = 3$ about the y-axis.



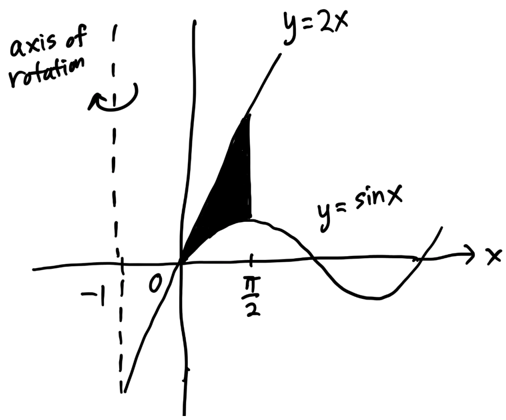
(ii)

$$\int_0^{\pi/2} 2\pi(x+1)(2x-\sin x)\,dx = \int_0^{\pi/2} \underbrace{2\pi(x+1)}_{\text{circumference}}\,\underbrace{(2x-\sin x)}_{\text{height}}\,dx.$$

The radius of the shell is $x+1 = x-(-1)$, so the solid is rotated about the line $x = -1$.

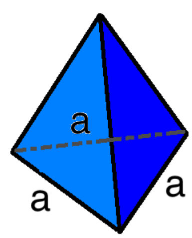
The height of the shell is $2x - \sin x$, so the solid is obtained by rotating the region bounded between $y = 2x$ (at the top) and $y = \sin x$ (below).

From the limits of integration, we conclude that this is the solid obtained by rotating the region abounded by the curves $y = 2x$ and $y = \sin x$, $0 \le x \le \pi/2$, about the line $x = -1$.

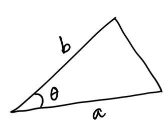


□

3. Find the volume of a pyramid with height h and base an equilateral triangle with side a.



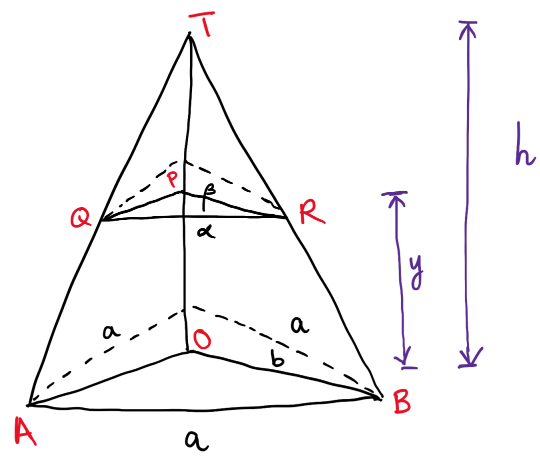
You can use the following sine formula for area of a triangle: Suppose θ is the angle between two sides (of length a and b respectively) of a triangle:



Then the area of the triangle is $\frac{1}{2}ab\sin\theta$. You can also use Heron's formula for the area of triangle.

Solution

Consider the cross-section at height y of the pyramid.



By definition, the volume of the pyramid is given by

$$V = \int_0^h A(y)\, dy,$$

where $A(y)$ is the area of the cross-section at height y.

The aim is to express the area $A(y)$ in terms of y and constants involving a, h.
The triangle at height y is similar to the triangle at the base. So each side of the triangle at height y is α and the angle between any two sides is $60°$ (it is an equilateral triangle). So using the sine formula for area, we have

$$A(y) = \frac{1}{2}\alpha\alpha \sin 60° = \frac{\sqrt{3}}{4}\alpha^2.$$

As shown in the figure above, the point O is the point on the base vertically below the point T. The vertical line OT cuts the cross-sectional triangle at height y at point P. The lengths of the sides AB and OB are denoted by a and b respectively, and the lengths of the sides QR and PR are denoted by α and β respectively.

- The triangles $\triangle PQR$ and $\triangle OAB$ are similar. This implies that

$$\frac{a}{b} = \frac{\alpha}{\beta} \tag{1}$$

- The triangles $\triangle PTR$ and $\triangle OTB$ are similar. This implies that

$$\frac{b}{h} = \frac{\beta}{h-y} \tag{2}$$

Combining (1) and (2), we have

$$\begin{aligned} \alpha &= \frac{a}{b} \cdot \beta = \frac{a}{b} \cdot \frac{b}{h}(h-y) & (3) \\ &= a\left(1 - \frac{y}{h}\right). & (4) \end{aligned}$$

Therefore, the required volume is

$$\begin{aligned} V &= \int_0^h A(y)\, dy = \int_0^h \frac{\sqrt{3}}{4}\alpha^2\, dy \\ &= \int_0^h \frac{\sqrt{3}}{4}a^2\left(1 - \frac{y}{h}\right)^2 dy \\ &= \frac{a^2\sqrt{3}}{4}\left[-\frac{h}{3}\left(1 - \frac{y}{h}\right)^3\right]_0^h = \frac{\sqrt{3}}{12}a^2 h. \end{aligned}$$

□

4. Evaluate the integral using integration by parts.

(i) $\int x \cos 5x\, dx$

(ii) $\int \ln \sqrt{x}\, dx$

(iii) $\int_0^1 (x^2+1)e^{-x}\, dx$

(iv) $\int_1^{\sqrt{3}} \tan^{-1}(1/x)\, dx$

(v) $\int_0^{1/2} \cos^{-1} x\, dx$

Solution

(i) Choose $u = x$, $v' = \cos 5x$. Then

$$u' = 1, \quad v = \frac{\sin 5x}{5}.$$

By the integration-by-parts formula,

$$\begin{aligned}\int x\cos 5x\,dx &= x\frac{\sin 5x}{5} - \int 1\cdot\frac{\sin 5x}{5}\,dx \\ &= x\frac{\sin 5x}{5} - \left(-\frac{\cos 5x}{25}\right) + C \\ &= \frac{x}{5}\sin 5x + \frac{1}{25}\cos 5x + C.\end{aligned}$$

(ii) Choose $u = \ln\sqrt{x}$, $v' = 1$. Then

$$u' = \frac{1}{\sqrt{x}}\frac{1}{2\sqrt{x}} = \frac{1}{2x}, \quad v = x.$$

By the integration-by-parts formula,

$$\begin{aligned}\int \ln\sqrt{x}\,dx &= x\ln\sqrt{x} - \int \frac{1}{2x}\cdot x\,dx \\ &= x\ln\sqrt{x} - \frac{x}{2} + C.\end{aligned}$$

(iii) Choose $u = x^2+1$, $v' = e^{-x}$. Then

$$u' = 2x, \quad v = -e^{-x}.$$

By the integration-by-parts formula,

$$\begin{aligned}\int_0^1 (x^2+1)e^{-x}\,dx &= \left[(x^2+1)(-e^{-x})\right]_0^1 - \int_0^1 2x(-e^{-x})\,dx \\ &= -2e^{-1} + 1 + \int_0^1 2xe^{-x}\,dx. \qquad (5)\end{aligned}$$

We will now apply the integration-by-parts formula to evaluate $\int_0^1 2xe^{-x}\,dx$. Choose $U = 2x$, $V' = e^{-x}$. Then

$$U' = 2, \quad V = -e^{-x}.$$

$$\begin{aligned}\int_0^1 2xe^{-x}\,dx &= \left[2x(-e^{-x})\right]_0^1 - \int_0^1 2(-e^{-x})\,dx \\ &= -2e^{-1} + 2\int_0^1 e^{-x}\,dx \\ &= -2e^{-1} + 2\left[-e^{-x}\right]_0^1 \\ &= -4e^{-1} + 2.\end{aligned}$$

Substituting this back into (5), we have

$$\int_0^1 (x^2+1)e^{-x}\,dx = -2e^{-1}+1-4e^{-1}+2 = -6e^{-1}+3.$$

(iv) Choose $u = \tan^{-1}(1/x)$, $v' = 1$. Then

$$u' = \frac{1}{1+(1/x)^2}\cdot\frac{d}{dx}(1/x) = \frac{1}{1+(1/x)^2}\cdot(-x^{-2}) = \frac{-1}{1+x^2}, \quad v = x.$$

By the integration-by-parts formula,

$$\begin{aligned}\int_1^{\sqrt{3}} \tan^{-1}(1/x)\,dx &= \left[x\tan^{-1}(1/x)\right]_1^{\sqrt{3}} - \int_1^{\sqrt{3}} \left(\frac{-1}{1+x^2}\cdot x\right)dx \\ &= (\sqrt{3}\tan^{-1}(1/\sqrt{3}) - \tan^{-1}1) + \int_1^{\sqrt{3}} \frac{x}{1+x^2}\,dx \\ &= \frac{\sqrt{3}\pi}{6} - \frac{\pi}{4} + \int_1^{\sqrt{3}} \frac{x}{1+x^2}\,dx.\end{aligned}$$

Substitute $u = 1+x^2$. Then $du = 2x\,dx \Longrightarrow xdx = \frac{1}{2}\,du$. Also, $x = 1 \Longrightarrow u = 1+x^2 = 1+1^2 = 2$, and $x = \sqrt{3} \Longrightarrow u = 1+x^2 = 1+\sqrt{3}^2 = 4$. So

$$\begin{aligned}\int_1^{\sqrt{3}} \frac{x}{1+x^2}\,dx &= \int_2^4 \frac{1}{2u}\,du \\ &= \frac{1}{2}\left[\ln|u|\right]_2^4 \\ &= \frac{1}{2}(\ln 4 - \ln 2) \\ &= \frac{\ln 2}{2}\end{aligned}$$

Hence,

$$\int_1^{\sqrt{3}} \tan^{-1}(1/x)\,dx = \frac{\sqrt{3}\pi}{6} - \frac{\pi}{4} + \frac{\ln 2}{2}.$$

(v) Choose $u = \cos^{-1}x$, $v' = 1$. Then

$$u' = -\frac{1}{\sqrt{1-x^2}}, \quad v = x.$$

By the integration-by-parts formula,

$$\begin{aligned}
\int_0^{1/2} \cos^{-1} x\,dx &= \left[x\cos^{-1} x\right]_0^{1/2} - \int_0^{1/2} -\frac{x}{\sqrt{1-x^2}}\,dx \\
&= \left(\frac{1}{2}\cos^{-1}\frac{1}{2} - 0\right) + \int_0^{1/2} \frac{x}{\sqrt{1-x^2}}\,dx \\
&= \frac{\pi}{6} + \int_0^{1/2} \frac{x}{\sqrt{1-x^2}}\,dx
\end{aligned}$$

Substitute $u = 1 - x^2$. Then $du = -2x\,dx \implies x\,dx = \left(-\frac{1}{2}\right) du$, and $x = 0 \implies u = 1$, $x = 1/2 \implies u = 3/4$. So

$$\begin{aligned}
\int_0^{1/2} \cos^{-1} x\,dx &= \frac{\pi}{6} + \int_1^{3/4} \frac{1}{u^{1/2}} \cdot \left(-\frac{1}{2}\right) du \\
&= \frac{\pi}{6} - \frac{1}{2}\int_1^{3/4} u^{-1/2}\,du \\
&= \frac{\pi}{6} - \frac{1}{2}\left[\frac{u^{1/2}}{1/2}\right]_1^{3/4} \\
&= \frac{\pi}{6} - ((3/4)^{1/2} - 1) \\
&= \frac{\pi}{6} - (\sqrt{3}/2 - 1) \\
&= \frac{1}{6}(\pi + 6 - 3\sqrt{3}).
\end{aligned}$$

□

5.

(a) Prove the reduction formula

$$\int \cos^n x\,dx = \frac{1}{n}\cos^{n-1} x \sin x + \frac{n-1}{n}\int \cos^{n-2} x\,dx.$$

(b) Use part (a) to evaluate $\displaystyle\int \cos^2 x\,dx$.

Solution

(a) Choose $u = \cos^{n-1} x$, $v' = \cos x$. Then

$$u' = (n-1)(\cos^{n-2} x)(-\sin x), \quad v = \sin x.$$

By the integration-by-parts formula,

$$\begin{aligned}
\int \cos^n x\,dx &= \cos^{n-1} x \sin x - \int (n-1)(\cos^{n-2} x)(-\sin x)(\sin x)\,dx \\
&= \cos^{n-1} x \sin x + \int (n-1)\cos^{n-2} x(1-\cos^2 x)\,dx \\
&= \cos^{n-1} x \sin x + \int (n-1)\cos^{n-2} x\,dx - (n-1)\int \cos^n x\,dx \\
n\int \cos^n x\,dx &= \cos^{n-1} x \sin x + \int (n-1)\cos^{n-2} x\,dx \\
\int \cos^n x\,dx &= \frac{1}{n}\cos^{n-1} x \sin x + \frac{n-1}{n}\int \cos^{n-2} x\,dx.
\end{aligned}$$

(b) Set $n = 2$. From part (a), we have

$$\begin{aligned}
\int \cos^2 x\,dx &= \frac{1}{2}\cos x \sin x + \frac{1}{2}\int 1\,dx \\
&= \frac{1}{2}\cos x \sin x + \frac{x}{2} + C.
\end{aligned}$$

□